

100 Top AI Engineer Interview Questions Based on Real Big Tech Patterns



AI ENGINEERING INSIDER

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Link: [ebook](#)

Coding, DSA, and AI Engineer Coding Screens

↳ **Q1.** In a 2D array, find the shortest path from the top-left cell to the bottom-right cell in a maze.

- ↳ **Q2.** How would you traverse a graph using BFS and DFS?
- ↳ **Q3.** Given a matrix, compute the sum of any rectangular submatrix efficiently.
- ↳ **Q4.** Given a string, find the length of the longest substring without repeating characters.
- ↳ **Q5.** Given a sorted 2D matrix, implement binary search to find a target value.
- ↳ **Q6.** Given a list of numbers, return the sorted order of only the odd numbers without using built-in sort.
- ↳ **Q7.** Given a string of parentheses, determine whether it is valid.
- ↳ **Q8.** Implement top-K frequent elements from a stream of user events.
- ↳ **Q9.** Design an LRU cache and explain time complexity.
- ↳ **Q10.** Given user click logs, group sessions by user and timestamp gap.
- ↳ **Q11.** Given a graph of users and friendships, find mutual friends efficiently.
- ↳ **Q12.** Implement a trie for autocomplete search.
- ↳ **Q13.** Given a sparse vector, implement dot product efficiently.
- ↳ **Q14.** Given a sparse matrix, design a memory-efficient representation and multiplication method.
- ↳ **Q15.** Given a stream of model predictions, compute rolling accuracy over a time window.

ML Coding and Algorithm Implementation

- ↳ **Q16.** Implement K-Means from scratch.
- ↳ **Q17.** Implement KNN classification from scratch.
- ↳ **Q18.** Implement logistic regression training using gradient descent.
- ↳ **Q19.** Implement AUC calculation from predicted scores and binary labels.
- ↳ **Q20.** Implement precision, recall, F1-score, and confusion matrix.
- ↳ **Q21.** Implement softmax and cross-entropy loss.
- ↳ **Q22.** Implement mini-batch gradient descent.
- ↳ **Q23.** Implement cosine similarity for embeddings.
- ↳ **Q24.** Implement TF-IDF vectorization.
- ↳ **Q25.** Implement a simple BM25 scoring function.
- ↳ **Q26.** Implement negative sampling for a recommendation model.
- ↳ **Q27.** Implement train/test split with stratification.
- ↳ **Q28.** Implement early stopping logic for model training.
- ↳ **Q29.** Implement model calibration using Platt scaling or isotonic regression.
- ↳ **Q30.** Implement a simple feature normalization pipeline.

ML Fundamentals and Evaluation

- ↳ **Q31.** Explain precision, recall, F1-score, and when each metric matters.
- ↳ **Q32.** Explain AUC-ROC and when it can be misleading.
- ↳ **Q33.** Explain bias-variance decomposition.
- ↳ **Q34.** What is TF-IDF, and why is it useful for text retrieval?
- ↳ **Q35.** What is perplexity, and how is it used to evaluate language models?
- ↳ **Q36.** What is calibration in machine learning?
- ↳ **Q37.** Explain logistic regression and how it differs from linear regression.
- ↳ **Q38.** Explain decision trees, random forests, and gradient boosting.
- ↳ **Q39.** How would you handle class imbalance in a fraud detection model?
- ↳ **Q40.** How would you choose between accuracy, AUC, log loss, and PR-AUC?
- ↳ **Q41.** How do you detect data leakage?
- ↳ **Q42.** How do you evaluate a model when labels are noisy?
- ↳ **Q43.** How do you evaluate crowdsourced labels?
- ↳ **Q44.** How would you design an offline evaluation framework before launching a model?
- ↳ **Q45.** How do you decide whether an offline metric improvement is meaningful enough for an A/B test?

Deep Learning and LLM Fundamentals

- ↳ **Q46.** Implement the LM head of a decoder-based language model.
- ↳ **Q47.** Why does layer normalization work better than batch normalization in transformers?
- ↳ **Q48.** How would you debug a model that is overfitting?
- ↳ **Q49.** Explain attention, self-attention, and multi-head attention.
- ↳ **Q50.** Explain positional embeddings and why transformers need them.
- ↳ **Q51.** What are the main flaws or limitations of current transformer architectures?
- ↳ **Q52.** Explain scaling laws and their practical limits.
- ↳ **Q53.** Explain pretraining, supervised fine-tuning, RLHF, DPO, and preference optimization.
- ↳ **Q54.** What is catastrophic forgetting, and how would you reduce it?
- ↳ **Q55.** Explain the difference between encoder-only, decoder-only, and encoder-decoder architectures.
- ↳ **Q56.** How do you reduce hallucination in LLM applications?
- ↳ **Q57.** How would you evaluate reasoning quality in an LLM?
- ↳ **Q58.** How would you evaluate the performance of a multilingual LLM?
- ↳ **Q59.** What is quantization, and how does it affect latency, memory, and accuracy?

↳ **Q60.** Why might int8 or int4 quantization not always improve real-world inference speed?

ML System Design

↳ **Q61.** Design a recommender system.

↳ **Q62.** Design a job recommendation system using user profiles, job profiles, and application history.

↳ **Q63.** Design a product recommendation system for e-commerce.

↳ **Q64.** Design a TikTok-style short-video recommendation system.

↳ **Q65.** Design a YouTube-style watch-time optimization system.

↳ **Q66.** Design an ads click-through-rate prediction system.

↳ **Q67.** Design a feed ranking system.

↳ **Q68.** Design a search ranking system with retrieval and ranking stages.

↳ **Q69.** Design a semantic search system.

↳ **Q70.** Design a visual image search system.

↳ **Q71.** Design a fraud detection system.

↳ **Q72.** Design an ETA prediction system for maps or delivery.

↳ **Q73.** Design a spam classifier for NLP.

↳ **Q74.** Design a content moderation classifier.

↳ **Q75.** Design a friend recommendation system.

- ↳ **Q76.** Design a multimodal search system.
- ↳ **Q77.** Design a real-time personalization system.
- ↳ **Q78.** Design a model serving platform for thousands of models.

GenAI, RAG, Agents, and LLMOps

- ↳ **Q79.** Design a RAG system for enterprise document search.
- ↳ **Q80.** How would you choose chunk size, overlap, and metadata strategy for RAG?
- ↳ **Q81.** How would you combine BM25 and vector search?
- ↳ **Q82.** How would you evaluate retrieval quality using Recall@K, MRR, and nDCG?
- ↳ **Q83.** How would you detect hallucinations in a RAG answer?
- ↳ **Q84.** How would you design citations and grounding for a customer-support chatbot?
- ↳ **Q85.** How would you build an AI agent that safely calls external tools?
- ↳ **Q86.** How would you prevent prompt injection in a tool-using agent?
- ↳ **Q87.** How would you design memory for an agentic AI system?
- ↳ **Q88.** How would you implement human-in-the-loop escalation?
- ↳ **Q89.** How would you evaluate an LLM agent end-to-end?
- ↳ **Q90.** How would you monitor latency, cost, hallucination rate, and tool failures in production?

↳ **Q91.** How would you design a scalable and efficient system for training a large language model under compute and data constraints?

↳ **Q92.** How would you deploy and monitor an LLM in production?

AI Safety, Alignment, and Responsible AI

↳ **Q93.** How would you design a language model that minimizes harmful outputs while remaining useful and expressive?

↳ **Q94.** How would you evaluate model bias?

↳ **Q95.** How would you test an LLM for jailbreak vulnerability?

↳ **Q96.** How would you design a safe content moderation pipeline?

↳ **Q97.** How would you handle privacy-sensitive training data?

↳ **Q98.** How would you prevent model extraction attacks?

↳ **Q99.** How would you decide whether an AI feature should be blocked, warned, escalated, or allowed?

↳ **Q100.** How would you balance safety, utility, latency, and business impact in a user-facing GenAI product?

Highest Priority for Big Tech AI Engineer Prep

↳ Practice **Q1, Q16, Q17, Q19, Q31–Q36, Q46–Q49, Q59–Q61, Q66–Q72, Q79–Q92, and Q93** first. These map most directly to the reported evidence above from Meta, Google, Amazon, Apple, OpenAI, Anthropic, Netflix, Microsoft, TikTok, Uber, LinkedIn, GitHub, and YouTube.